

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

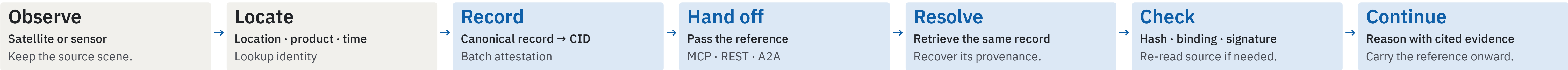
EMEM gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence.

A compact citation keeps the observation traceable across agents and time.

Jaya Kumari · Avijeet Singh · Vortex AI · avijeet@vortex.ai
Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)



From observation to the next agent



1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a source read. A relay changes the evidence. Agent B acts on it or checks the reference.



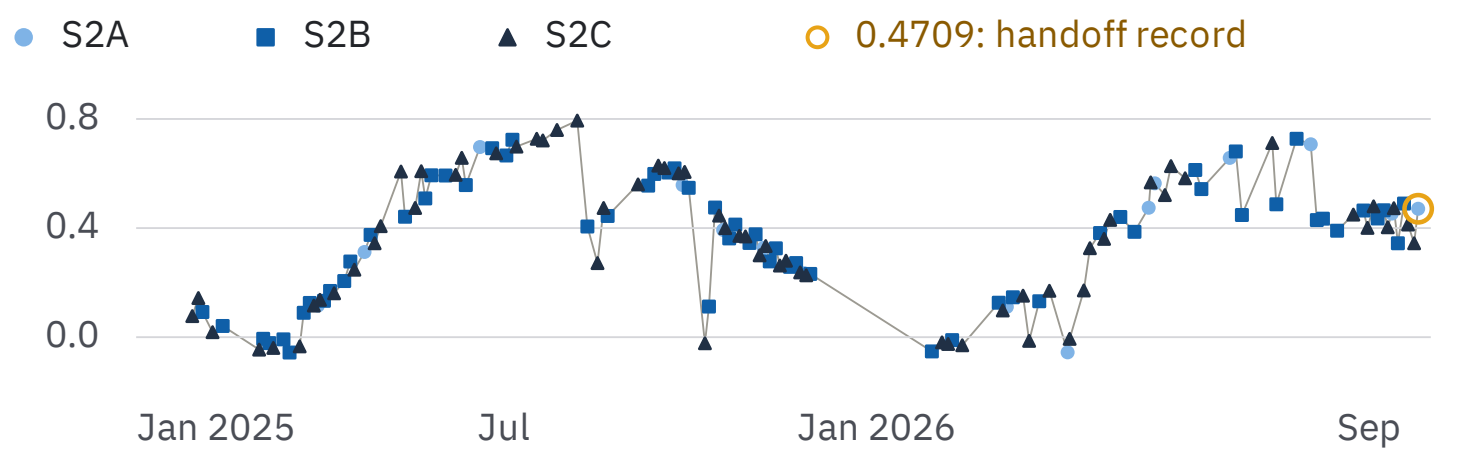
In the controlled handoff test, prose led agents to act on 254 of 276 corruptions; a reference with explicit checking instructions, 0 of 300.

2 Can a time series retain its evidence?

NDVI history, with citations

Each value retains its dated record.

KEYLONG · SENTINEL-2 L2A · 141 RECORDS



Cloud/snow screening and processing harmonisation were not applied; physical interpretation needs both.

3 Where did the evidence change?

A changed value needs a cause

A cited value can change during a handoff.

Between acquisitions, the surface, sensor or processing may also change.

$$\Delta z = \Delta_{\text{env}} + \Delta_{\text{sensor}} + \Delta_{\text{geo}} + \Delta_{\text{encoder}} + \epsilon$$

Surface change, instrument, registration, encoder and residual. Δz is the later value minus the earlier one. The attribution ledger links evidence to each term; the numeric split remains open.

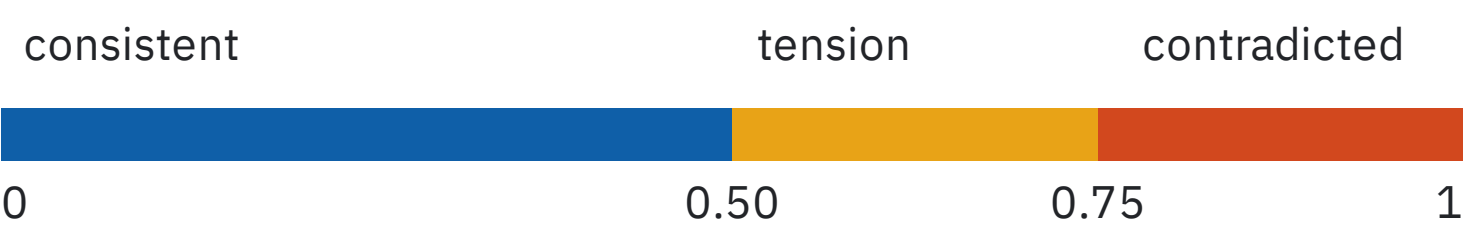
IMPLEMENTED · DRIFT-ANCHOR SCORE

$$d = |x - a|$$
$$r = d / (3\sigma); s = r / (1 + r)$$

for finite x , a and $0 < \sigma < \infty$

Invalid σ : $s = 0$ if $d = 0$; otherwise $s = 1$.

x : output · a : anchor · σ : anchor uncertainty



Tests pin $3\sigma \rightarrow 0.50$ and $9\sigma \rightarrow 0.75$.

Score boundaries: consistent < 0.50 ; tension < 0.75 .

The implemented anchor score measures disagreement, not its cause. SAT-042 exercises it in a reference harness; live device-to-anchor wiring remains next work.

Use it in an EO workflow

Compare dates · recall, diff

Assemble evidence · merge, trace

Retain conflicts · competing, evolve

Refresh inputs · ensure, valid

Compare vegetation observations, keep conflicting product estimates and pass the cited evidence onward. Single-hop retrieval is implemented; multi-hop planning remains open.
20 upstream offline SDK / encoding tests passed.

Model outputs are records too

Satellite input → encoder → vector record → content address

Prithvi 1,024 values

checkpoint digest inside the record

TESSERA 128 values

product year 2024; no checkpoint digest

Archived vectors: Prithvi names a checkpoint digest; TESSERA names a product year. These encoders are retired on the deployment; stored records remain addressable.

THE CONTRIBUTION

Checkable observation handoff

An agent passes a reference to a specific physical observation; the receiver resolves it and checks the evidence.

Built with standard BLAKE3, Ed25519, CBOR and Merkle logs. STAC, openEO, PROV, C2PA, RAG and temporal storage provide complementary layers.

11 Use emem. Carry the evidence forward.

Measured separately: same record and value through 11 client paths; receipt signatures checked on 9 (30 Sep 2026).

ChatGPT @emem plugin Enable emem; paste a token.	Claude Code plugin / MCP Resolve from chat or terminal.	Visual Studio Code MCP connector Copilot agent mode: @mcp emem	Dify Marketplace plugin Referent Lock / EU DR workflows	Salesforce MuleSoft Exchange listing Open Vortex AI MCP Server	CONNECT
MORE CLIENTS Cursor Gemini CLI	PROTOCOLS MCP · A2A REST / OpenAPI	DEVELOPER Python · TypeScript Docker · LangGraph memory GitHub source	FRAMEWORK EXAMPLES LangChain · LlamaIndex AutoGen · CrewAI · Mastra Agno (see setup notes)	DISCOVERY / MIRRORS GitHub MCP · MCP Registry Glama · Smithery · ClawHub Context7 · Hugging Face (older mirror)	Setup · EO tasks Reproduce

Implementation: emem 8e9b401. Measurements: 29 Sep to 1 Oct 2026. · The linked methods connect this poster's figures to their records, source data and reproducible checks. · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)